

$\mathcal{M} =$

$$(2\pi)^4 \bar{v}^{(s_2)}(p_2)\, ige\gamma^\mu\, u^{(s_1)}(p_1)\, \frac{-ig_{\mu\nu}}{q^2}\, \bar{u}^{(s_3)}(p_3)\, ige\gamma^\mu\, v^{(s_4)}(p_4)\, \delta(p_1+p_2-q)$$

$$\delta(q-p_2-p_4)\cdot dq$$

$$= (2\pi)^4 \left[-\frac{ge^2}{(p_1+p_2)^2}\right] [\bar{v}(2)\gamma^\mu u(1)][\bar{u}(3)\gamma_\mu v(4)]\, dq^4$$

Casimir trick :

$$|\bar{v}(2)\gamma^\mu u(1)|^2 = [(\bar{v}(2)\gamma^\mu u(1)][\bar{v}(2)\gamma^\mu u(1)]^*$$

$$= \mathrm{Tr}[\gamma^\mu(\not{p}_2 + m_e c)\gamma^\nu(\not{p}_1 + m_e c)]$$

$$= \mathrm{Tr}[\gamma^\mu \not{p}_2 \gamma^\nu \not{p}_1] + m_e c[\mathrm{Tr}(\gamma^\mu \not{p}_1 \gamma^\nu) + \mathrm{Tr}(\gamma^\mu \not{p}_2 \gamma^\nu)]$$
$$+ m_e^2 c^2 \mathrm{Tr}(\gamma^\mu \gamma^\nu)$$

$$\mathrm{Tr}(\gamma^\mu \not{p}_2 \gamma^\nu \not{p}_1) = (p_2)_\lambda (p_1)_\sigma \mathrm{Tr}(\gamma^\mu \gamma^\lambda \gamma^\nu \gamma^\sigma)$$

$$= (p_2)_\lambda (p_1)_\sigma 4($$

$$|\bar{u}(3)\gamma_\mu v(4)|^2 = \mathrm{Tr}[\gamma_\mu(\not{p}_3 + m_\nu c)\gamma_\nu(\not{p}_4 + m_\nu c)]$$